

Realizability and minimum dimension of Fano half-rank profiles over $\mathbb{F}_2$

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Abstract

Let V be a finite-dimensional vector space over $\mathbb{F}_2$ and let

$$B : \mathbb{F}_2^3 \longrightarrow \text{Alt}(V)$$

be a linear family of alternating bilinear forms. Its half-rank profile is the integer-valued function

$$h(a) = \frac{1}{2} \text{rank}(B_a), \quad a \in \mathbb{F}_2^3 \setminus \{0\}.$$

We classify these labelled seven-point profiles in two senses. First, we determine exactly which nonnegative integer functions on the Fano plane occur in some finite dimension. The triangle inequalities are sufficient except for seven primitive line jumps and seven infinite parity cosets supported on singleton faces. Second, for every realizable profile h we determine the minimum possible dimension $\mu(h)$ of V . If $t = \max h$, then $\mu(h) \in \{2t, 2t + 1\}$, and we give a complete description of the profiles that require the extra dimension: two height-one orbits, three height-two orbits, and one explicit infinite family E_t .

The nonrealizability arguments use Pfaffian parity, a simultaneous decomposition of commuting self-adjoint idempotents, and a rank-two Pfaffian update identity. The sufficiency proofs are computer assisted only at finite, exact interfaces: an integral Hilbert-basis certificate and a conductor reduction to a verified collection of profiles of height at most six.

1 Introduction

Rank distributions of linear spaces of alternating matrices arise in finite geometry, coding theory, and the study of bilinear and quadratic forms. The two-dimensional parameter case asks for the ranks of a triple $(A, B, A + B)$. It has been studied in a precise enumerative form, for example by Pott, Schmidt, and Zhou [1]. Passing from a two-dimensional parameter space to a three-dimensional one replaces the three nonzero parameters by the seven points of the Fano plane. The resulting seven ranks are not independent: each Fano line is a two-dimensional subfamily and therefore satisfies rank subadditivity.

The purpose of this paper is to answer two natural questions for this first seven-point case over $\mathbb{F}_2$.

1. Which labelled seven-tuples occur as the half-ranks of a linear net of alternating forms in some finite dimension?
2. For a realizable profile, what is the least dimension in which it occurs?

The answer to the first question is almost, but not quite, the integral metric cone of the Fano plane. There are two kinds of holes. One is finite: a primitive profile that jumps from one off a line to two on the line. The other consists of seven infinite parity cosets on four-generated faces of the cone. Both obstructions have structural explanations rather than being artifacts of a bounded search.

The answer to the second question is unexpectedly rigid. The evident lower bound is twice the largest half-rank. Every realizable profile meets this bound or misses it by exactly one. The profiles that miss it form a short finite list in heights one and two and a single infinite family obtained by repeatedly adding one Fano cut.

Our classification is a classification of labelled rank profiles, not of alternating-matrix spaces up to isomorphism. Eisenbud and Koh classified three-dimensional spaces of 4×4 and 5×5 alternating matrices over an algebraically closed field [2]. Gow obtained dimension bounds for spaces with restricted rank sets over finite fields [3], while Qiao studied the enumeration of alternating-matrix spaces with explicit coordinates [4]. These results are closely related in language but have a different output. Here the field is fixed to $\mathbb{F}_2$, the matrix size is arbitrary, and the output is a complete seven-point realizability and minimum-dimension criterion.

The proof has two finite exact components. We certify the Hilbert basis of the Fano triangle cone and later certify a finite conductor for sharp-dimensional realizability. Both certificates use integer or binary arithmetic; their role and logical boundary are stated explicitly in Sections 3, 7, and 8.

2 Definitions and basic properties

Throughout, V is a finite-dimensional vector space over $\mathbb{F}_2$. Write $\text{Alt}(V)$ for the vector space of alternating bilinear forms on V . Equivalently, after choosing a basis, its elements are symmetric matrices with zero diagonal. Their ranks are even. We use $\mathbb{N} = \{0, 1, 2, \dots\}$.

Let

$$P = \mathbb{F}_2^3 \setminus \{0\}.$$

The seven elements of P are the labelled points of the Fano plane. Its lines are the triples

$$\{a, b, a + b\}, \quad a, b \in P, \ a \neq b.$$

An alternating net is a linear map

$$B : \mathbb{F}_2^3 \longrightarrow \text{Alt}(V).$$

Its half-rank profile is the function $h_B : P \rightarrow \mathbb{N}$ given by

$$h_B(a) = \frac{1}{2} \text{rank}(B_a). \tag{2.1}$$

We call h realizable if $h = h_B$ for some V and some alternating net B . The realizable profiles form an additive semigroup: orthogonal direct sum of nets adds profiles coordinatewise.

For a Fano line L define the cut profile c_L and the line jump ℓ_L by

$$c_L(x) = \begin{cases} 0, & x \in L, \\ 1, & x \notin L, \end{cases} \quad \ell_L(x) = \begin{cases} 2, & x \in L, \\ 1, & x \notin L. \end{cases} \quad (2.2)$$

For $p \in P$, define the singleton jump

$$s_p(x) = \begin{cases} 2, & x = p, \\ 1, & x \neq p. \end{cases} \quad (2.3)$$

Rank subadditivity gives, on every Fano line,

$$h(a + b) \leq h(a) + h(b). \quad (2.4)$$

Let $C_{\mathbb{Z}}$ be the semigroup of all nonnegative integer functions on P satisfying the three versions of (2.4) on each line. We call it the integral Fano triangle cone.

The effective dimension of a realizable profile is

$$\mu(h) = \min\{\dim V : h = h_B \text{ for a net on } V\}. \quad (2.5)$$

Adding a common radical does not change the profile. Consequently, once h is realizable in dimension $\mu(h)$, it is realizable in every dimension at least $\mu(h)$.

3 The integral Fano triangle cone

The cone $C_{\mathbb{Z}}$ provides the necessary inequalities and the finite additive framework used in the realizability proof.

Lemma 3.1 (extreme rays and Hilbert basis). *The rational cone underlying $C_{\mathbb{Z}}$ has fourteen primitive extreme rays: the seven cut profiles c_L and the seven line jumps ℓ_L . Its integral Hilbert basis has 78 elements. Of these, 64 are realizable in dimension at most four. The remaining fourteen are precisely the seven s_p and the seven ℓ_L .*

Exact certificate

There are 28 defining inequalities: seven nonnegativity inequalities and three triangle inequalities on each of seven lines. The program `verify_fano_triangle_extreme_rays.cpp` examines every six-subset of the 28 inequalities, computes its generalized cross product over the integers, and retains the primitive feasible rays. It finds exactly the fourteen profiles above.

The program `verify_fano_triangle_hilbert_basis.py` reconstructs the facet incidences, forms an exact pulling triangulation with twelve simplicial cones, and enumerates their half-open fundamental parallelepipeds. Every determinant, adjugate, divisibility test, and decomposability test is exact. The union gives the stated 78-element Hilbert basis. The alternating-net witnesses for the realizable atoms are reconstructed and checked over $\mathbb{F}_2$ by `verify_realized_strengthened_fano_atoms.py`.

The lemma is therefore a finite computer-assisted lemma. It does not infer a general statement from a numerical bound.

Lemma 3.2 (multiples of missing atoms). *For every point p and every line L ,*

$$2s_p = \sum_{K:p \notin K} c_K. \quad (3.1)$$

Moreover, $m\ell_L$ is realizable for every $m \geq 2$.

Proof. There are four Fano lines avoiding p , and direct evaluation gives (3.1). Each c_K is realized on a two-dimensional symplectic space, so $2s_p$ is realizable by orthogonal direct sum. For the line jump, explicit matrix spaces give

$$2\ell_L = (4, 4, 4, 2, 2, 2, 2), \quad 3\ell_L = (6, 6, 6, 3, 3, 3, 3), \quad (3.2)$$

after a suitable labelling with L in the first three coordinates. The matrices and all seven ranks are checked in `verify_fano_43_counterexample.py`. Every integer $m \geq 2$ is a nonnegative combination of 2 and 3, so orthogonal direct sums finish the proof. $\square$

4 The realizability semigroup

We now identify the two genuine boundaries left by Lemma 3.1.

Lemma 4.1 (rank-additive reduction). *Let $A, C : V \rightarrow V^*$ be linear maps. If*

$$\text{rank}(A + C) = \text{rank}(A) + \text{rank}(C), \quad (4.1)$$

then

$$\text{im}(A) \cap \text{im}(C) = 0, \quad \ker(A + C) = \ker(A) \cap \ker(C). \quad (4.2)$$

Proof. The image of $A + C$ is contained in $\text{im}(A) + \text{im}(C)$. Equality of dimensions in (4.1) forces the latter sum to be direct and forces $\text{im}(A + C)$ to equal it. If $(A + C)x = 0$, then $Ax = Cx$ lies in the intersection of the two images and hence both terms vanish. The reverse kernel inclusion is immediate. $\square$

Proposition 4.2 (primitive line jumps are not realizable). *For every Fano line L , the profile ℓ_L is not realizable.*

Proof. Write $L = \{u, v, u + v\}$ and choose $w \notin L$. Suppose ℓ_L were realized. Then $P = B_u$ has rank four, while

$$B_w, B_{w+u}, B_{w+v}, B_{w+u+v}$$

all have rank two. Both decompositions

$$P = B_w + B_{w+u} = B_{w+v} + B_{w+u+v} \quad (4.3)$$

are rank additive. Lemma 4.1 shows that $\text{rad}(P)$ lies in the radical of each of the four rank-two forms. The other two forms on L are sums of these forms, so the entire net descends to $V/\text{rad}(P)$, which has dimension four.

On a four-dimensional space an alternating form is nondegenerate exactly when its maximal Pfaffian is nonzero. The Pfaffian of the net is a zero-constant Boolean polynomial of degree at most two in the three parameters. Every such function has even Hamming weight on $\mathbb{F}_2^3$: each monomial of degree less than three has even total weight, and the value at zero is zero. The profile ℓ_L , however, requires the Pfaffian to be nonzero exactly on the three points of L . This odd weight is impossible. $\square$

Proposition 4.3 (the singleton face). *Fix $p \in P$. Suppose a realizable profile h satisfies equality in all three triangle inequalities whose opposite point is p ; equivalently,*

$$h(p) = h(a) + h(a + p) \quad (4.4)$$

for the three unordered pairs $\{a, a + p\}$ in $P \setminus \{p\}$. Then

$$h \in \mathbb{N}\{c_L : p \notin L\}. \quad (4.5)$$

Conversely, every profile in the semigroup on the right is realizable and satisfies (4.4).

Proof. Let $P_0 = B_p$. Lemma 4.1 applied to each equality in (4.4) shows that all forms descend to $V/\text{rad}(P_0)$. We may therefore suppose that P_0 is nondegenerate.

Choose two other parameter directions and write the corresponding forms as

$$A = P_0 T, \quad C = P_0 S.$$

We use the following elementary observation. If an endomorphism T satisfies

$$\text{rank}(T) + \text{rank}(I + T) = \dim V, \quad (4.6)$$

then T is idempotent. Indeed, $\text{im}(T) + \text{im}(I + T) = V$, so (4.6) makes the sum direct. The vector $(I + T)Tx = T(I + T)x$ lies in both summands and must vanish.

The three equalities (4.4) imply that T , S , and $T + S$ satisfy (4.6), hence are idempotent. Alternation of A and C says that T and S are self-adjoint with respect to P_0 . In characteristic two,

$$(T + S)^2 = T + S$$

together with $T^2 = T$ and $S^2 = S$ gives $TS = ST$.

Thus V is the direct sum of the four simultaneous eigenspaces V_{ij} for (T, S) . Self-adjointness makes distinct eigenspaces orthogonal for P_0 . Since P_0 is nondegenerate, its restriction to each V_{ij} is nondegenerate; consequently every V_{ij} has even dimension.

On V_{ij} each parameter form is a scalar linear function on $\mathbb{F}_2^3$ times the fixed symplectic form $P_0|_{V_{ij}}$. That scalar is one at p , so its zero set is a line avoiding p . If $\dim V_{ij} = 2m_{ij}$, this block contributes m_{ij} times the corresponding cut profile. Summing the four blocks proves (4.5). The converse follows from the standard two-dimensional realizations of the cuts. $\square$

Corollary 4.4 (singleton parity cosets are holes). *For every $p \in P$, no profile in*

$$s_p + \mathbb{N}\{c_L : p \notin L\} \tag{4.7}$$

is realizable.

Proof. All profiles in (4.7) satisfy the equalities (4.4). The four cut vectors in (4.5) are linearly independent on that face. Identity (3.1) says that s_p is the nontrivial order-two lattice coset represented by half their sum. Hence the coset (4.7) is disjoint from the semigroup (4.5), while Proposition 4.3 places every realizable profile on the face in (4.5). $\square$

The remaining sufficiency step is finite once the Hilbert basis is known.

Lemma 4.5 (finite coupling certificate). *Let H be the 78-element Hilbert basis from Lemma 3.1, and let*

$$M = \{s_p : p \in P\} \cup \{\ell_L : L \text{ a Fano line}\}$$

be its fourteen nonrealizable atoms. There is a finite exact witness library with the following properties.

1. *Every sum of two atoms in M is realizable.*
2. *Every sum of three atoms in M is realizable, except $3s_p$.*
3. *The sum of any ℓ_L and any atom of $H \setminus M$ is realizable.*
4. *The sum $s_p + g$, for $g \in H \setminus M$, is nonrealizable exactly when $g = c_L$ for a line L avoiding p .*

Exact certificate

The program `verify_fano_profile_two_saturation.py` reconstructs 218 profiles with separately checked alternating-net witnesses, closes them additively in the finite target box, and verifies all 105 unordered pairs, 560 unordered triples, and 896 mixed targets. It reports precisely the four rules above. All profile additions and rank checks are exact.

Theorem 4.6 (realizability classification). *A profile $h : P \rightarrow \mathbb{N}$ is realizable in some finite dimension if and only if $h \in C_{\mathbb{Z}}$ and neither of the following holds:*

1. $h = \ell_L$ for a Fano line L ;
2. for some $p \in P$,

$$h \in s_p + \mathbb{N}\{c_L : p \notin L\}. \quad (4.8)$$

Proof. Necessity of the triangle inequalities follows from rank subadditivity. Proposition 4.2 and Corollary 4.4 prove necessity of the two exclusions.

Conversely, decompose h into elements of the Hilbert basis H . The elements of $H \setminus M$ are realizable. It remains to absorb the multiset of missing atoms.

If its size is even, pair the missing atoms and use Lemma 4.5(1). If its size is odd and at least three, a mixed triple can first be realized by Lemma 4.5(2), after which the rest can be paired. The only unmixed cases require separate attention. Repeated line jumps are realizable from multiplicity two onward by Lemma 3.2. Repeated singleton jumps are realizable in even multiplicity by (3.1), while an odd multiplicity reduces to one s_p plus cuts avoiding p ; this is exactly the excluded coset (4.8).

Finally, if one singleton atom remains together with good Hilbert atoms, Lemma 4.5(4) says that failure occurs exactly when all additions preserve the same singleton face, again giving (4.8). A remaining line jump is absorbed by Lemma 4.5(3), unless it is the primitive line jump itself. Thus every profile not in the two excluded families is realizable. $\square$

Regression check

As a destructive check, `probe_fano_profile_semigroup_holes.py` enumerates all 61,075 points of $C_{\mathbb{Z}}$ with $\max h \leq 6$. The witness semigroup contains 60,578 of them. The remaining 497 profiles agree point for point with the two families in Theorem 4.6. This bounded enumeration is not used to infer the theorem.

5 Minimum effective dimension

For a nonzero realizable profile h , set

$$t = \max_{a \in P} h(a). \quad (5.1)$$

Every realization has dimension at least $2t$. We now classify exactly when one additional dimension is necessary.

Choose linearly independent points $a, b, c \in P$, and put $s = a + b + c$. For $t \geq 3$, define $E_{t;a,b,c}$ by

$$E_{t;a,b,c}(x) = \begin{cases} t, & x \in \{a, b, c\}, \\ t - 1, & x = s, \\ 1, & x \in \{a + b, a + c, b + c\}. \end{cases} \quad (5.2)$$

The group $\mathrm{GL}(3, 2)$ has one orbit of these profiles, of labelled size 28 for each t .

Theorem 5.1 (minimum-dimension classification). *Let h be a nonzero realizable profile and let $t = \max h$. Then*

$$\mu(h) = 2t + \varepsilon(h), \quad \varepsilon(h) \in \{0, 1\}. \quad (5.3)$$

The value $\varepsilon(h) = 1$ occurs exactly in the following cases.

1. **Height one.** *The profile is constant one, or it has one zero coordinate and six coordinates equal to one.*
2. **Height two.** *Every coordinate is positive, and the set $\{x : h(x) = 2\}$ is a noncollinear three-set, the complement of a two-set, or all seven points.*
3. **The infinite family.** *The profile is $E_{t;a,b,c}$ for some $t \geq 3$ and some linearly independent a, b, c .*

The zero profile satisfies $\mu(0) = 0$.

We prove necessity of the three exceptional families first.

Proposition 5.2 (height one). *The two height-one profiles in Theorem 5.1 have minimum dimension three. Every other realizable height-one profile is a cut and has minimum dimension two.*

Proof. On a two-dimensional space, $\text{Alt}(V)$ is one-dimensional. The set of parameters giving the unique nonzero form is the complement of the kernel of a linear functional, hence is the complement of a Fano line. These are exactly the cuts.

On a three-dimensional space every nonzero alternating form has rank two. A rank-two linear map from $\mathbb{F}_2^3$ to $\text{Alt}(\mathbb{F}_2^3)$ with a prescribed one-dimensional kernel gives the single-zero profile, while an isomorphism gives the constant-one profile. $\square$

Proposition 5.3 (height two). *The three height-two orbits in Theorem 5.1 have minimum dimension five.*

Proof. In dimension four, the set of full-rank directions is the support on P of the maximal Pfaffian, a zero-constant Boolean polynomial of degree at most two. As in Proposition 4.2, its weight is even. The three stated orbits have, respectively, 3, 5, and 7 full-rank directions, so none occurs in dimension four.

Exact five-dimensional alternating-net witnesses exist for every member of the three orbits. They are included in the verified low-layer construction library; `probe_n5_profile_saturation.cpp` reconstructs the complete target set and checks the ranks over $\mathbb{F}_2$. Hence the minimum is five. $\square$

Lemma 5.4 (rank-two Pfaffian update). *Let A and X be alternating forms on a $2t$ -dimensional space. For fixed A , there is a linear functional λ_A on $\text{Alt}(V)$ such that*

$$\text{Pf}(A + X) = \text{Pf}(A) + \lambda_A(X) \quad (5.4)$$

whenever $\text{rank}(X) \leq 2$.

Proof. The principal-minor expansion of the Pfaffian is

$$\text{Pf}(A + X) = \sum_{I \subseteq [2t], |I| \text{ even}} \text{Pf}(X_I) \text{Pf}(A_{I^c}). \quad (5.5)$$

If $\text{rank}(X) \leq 2$, then every principal submatrix X_I of order at least four is singular, so its Pfaffian vanishes. Only the empty set and the two-element subsets remain. The empty set gives $\text{Pf}(A)$, and the two-element terms are linear in the entries of X . Their sum is a single linear functional λ_A that works simultaneously for every rank-two X . $\square$

Proposition 5.5 (the family E_t). *For every $t \geq 3$,*

$$\mu(E_{t;a,b,c}) = 2t + 1. \quad (5.6)$$

Proof. Suppose first that E_t were realized in dimension $2t$. Use a full-rank direction as an origin and write the forms in three full-rank directions as

$$A, \quad A + R, \quad A + Q.$$

The three pair-sum directions in (5.2) have rank two, so $R, Q, R + Q$ all have rank two. Over $\mathbb{F}_2$, a nondegenerate alternating form has Pfaffian one. Lemma 5.4 and nondegeneracy of $A + R$ and $A + Q$ imply

$$\lambda_A(R) = \lambda_A(Q) = 0.$$

Linearity gives $\lambda_A(R + Q) = 0$, and hence $A + R + Q$ is also nondegenerate. But the direction s in (5.2) is required to have rank $2t - 2$. This is a contradiction, so $\mu(E_t) \geq 2t + 1$.

An exact seven-dimensional witness realizes E_3 . Let

$$L = \{a + b, a + c, b + c\}.$$

Then coordinatewise

$$E_{t+1} = E_t + c_L. \quad (5.7)$$

Taking an orthogonal direct sum with the two-dimensional realization of c_L therefore realizes E_t in dimension $2t + 1$ for every $t \geq 3$. $\square$

6 Sharp realizations and the conductor theorem

Call a realizable profile h **sharp** if it is realizable in dimension $2 \max h$; in particular, the zero profile is sharp in dimension zero. Propositions 5.2, 5.3, and 5.5 show that the listed exceptional profiles are not sharp. To prove that there are no others, we reduce every height to a finite exact calculation.

By the action of $\mathrm{GL}(3, 2)$, choose an anchor point labelled 1 at which h attains its maximum. Consider the chamber

$$C_1 = \{h \in C_{\mathbb{Z}} : h(1) \geq h(x) \text{ for all } x \in P\}. \quad (6.1)$$

Lemma 6.1 (sharp chamber generators). *The cone C_1 has 30 primitive rays. Replace them by the following scaled profiles:*

- *leave each cut ray unchanged;*
- *multiply each of the six anchored single-zero height-one rays by two;*
- *multiply the constant height-one ray by three;*
- *multiply each primitive line-jump ray by two;*
- *leave every other primitive ray unchanged.*

Every resulting profile is sharp, and all have the same anchor as a maximum coordinate. Consequently, every nonnegative integer sum of them is sharp.

Proof. The cut constructions are two-dimensional. For the two exceptional scaled height-one rays, explicit sharp witnesses are required rather than direct sums of the three-dimensional realizations in Proposition 5.2. In dimension four, the alternating edge-mask triple $(12, 0, 22)$ has profile

$$(2, 0, 2, 2, 2, 2, 2),$$

and Fano relabelling gives all doubled single-zero rays. In dimension six, let $K = \mathbb{F}_8$ and put $V = K \oplus K$. The trace-pairing forms

$$B_a((x, y), (x', y')) = \mathrm{Tr}_{K/\mathbb{F}_2}(a(xy' + x'y))$$

are nondegenerate for every $a \neq 0$. Restricting a to a three-dimensional $\mathbb{F}_2$ -basis of K gives the constant profile $(3, 3, 3, 3, 3, 3, 3)$. Both constructions are reconstructed by `verify_scaled_chamber_ray_witnesses.py`.

Lemma 3.2 gives sharp realizations of doubled line jumps. The remaining rays have exact sharp witnesses in the low-layer library.

For orthogonal direct sums, profiles add and dimensions add. Since the chosen anchor is maximal in every summand, its coordinate equals the sum of the individual maxima. Thus the resulting dimension is twice the maximum of the sum profile. $\square$

The factor three for the constant ray is essential: the constant-two profile is one of the height-two exceptions.

Lemma 6.2 (finite conductor certificate). *Every lattice point of C_1 is exactly one of the following:*

1. *a sharp profile;*
2. *a nonrealizable profile excluded by Theorem 4.6;*
3. *one of the height-one or height-two exceptions of Theorem 5.1;*
4. *a member of the family E_t .*

Exact certificate and reduction

The completeness of the 30 primitive chamber rays is independently certified before the conductor calculation. The program `verify_fano_anchor_cone_extreme_rays.cpp` constructs all 34 inequalities (nonnegativity, Fano triangles, and six anchor inequalities), examines all 1,344,904 six-subsets, and proves that their primitive feasible null rays are exactly the 30 rays used by the conductor.

The 30 scaled rays in Lemma 6.1 then have a pulling triangulation into 81 full-dimensional simplicial cones. Their determinant distribution is

$$\{8 : 24, 16 : 36, 24 : 18, 48 : 3\}. \quad (6.2)$$

The union of their half-open fundamental parallelepipeds contains 226 integer residue profiles, all of height at most six. In any fixed simplicial cone, every lattice point has a unique decomposition

$$h = r + \sum_i m_i g_i, \quad m_i \in \mathbb{N}, \quad (6.3)$$

where r is the residue and the g_i are the seven scaled rays of that cone.

The exact transition table has five kinds of residues. A sharp residue remains sharp after additions. A height-two exception is exceptional only with no addition; the first addition reaches a verified sharp base. A height-one exception either becomes sharp, passes once through a height-two exception, or follows one repeated cut and generates $E_3, E_4, \dots$. A nonrealizable singleton-face residue either stays in its forbidden coset or leaves the face and reaches a sharp base. The zero residue is generated directly by the rays.

All resulting sharp bases have height at most six. There are 812 distinct such bases. `verify_fano_graded_conductor.py` certifies the 81 cones, 226 residues, and every transition using exact integer arithmetic. The normalized list of 812 bases is frozen in `verification/results/required_sharp_bases.tsv`; `verify_fano_graded_required_bases.py` regenerates the conductor output and checks exact set equality with that file.

The witness side is fail closed. Complete five- and six-dimensional low layers are reconstructed by `probe_n5_profile_saturation.cpp` and `probe_n6_profile_saturation.cpp`; the exceptional E_3 witness is checked by `probe_n7_noncollinear_full_exception.cpp`. The dimension-eight boundary is constructed by exact binary-rank searches plus a fixed targeted witness. The remaining height-five and height-six representatives are checked by the finite-field trace and explicit matrix scripts in the verification package. Every search uses a fixed seed and exits nonzero unless its entire stated target set is covered.

Finally, `verify_fano_graded_excess_bound.cpp` reconstructs the complete low-height chamber from these witness-certified generators and checks every one of the frozen 812 bases at sharp dimension. It reports

```
REFINED_CLASSIFICATION errors=0
REQUIRED_BASE_COVER bases=812 errors=0
```

Randomized routines were used only to discover some matrices. They have fixed seeds, and every retained matrix is checked by exact Gaussian elimination. A failure to find a matrix is never used as a proof of nonrealizability.

Proof of Theorem 5.1. The lower bound $\mu(h) \geq 2t$ follows from the definition of t . The zero profile is realized on the zero space. Propositions 5.2, 5.3, and 5.5 prove that every listed exception has minimum dimension $2t + 1$.

For any other realizable profile, apply a Fano automorphism so that it lies in C_1 . Lemma 6.2 says it is sharp. Therefore it is realized in dimension $2t$. Undoing the relabelling preserves dimension and proves (5.3) for every realizable profile. $\square$

7 Computational proof boundary

The two main theorems combine ordinary proofs with finite exact certificates. For clarity, the logical division is as follows.

General arguments independent of computation

- rank subadditivity and the triangle inequalities;
- nonrealizability of primitive line jumps by four-dimensional descent and

Pfaffian parity;

- the exact singleton-face semigroup by commuting self-adjoint idempotents;
- the lower bounds for the height-one and height-two exceptions;
- the arbitrary-dimensional rank-two Pfaffian update identity;
- nonsharpness of every E_t and its propagation by Fano cuts;
- orthogonal-sum propagation once a finite witness is available.

Finite exact certificates

- completeness of the fourteen extreme rays and the 78-element Hilbert basis;
- realization of the finite atom and coupling libraries;
- completeness of the 30 primitive rays of the anchored chamber;
- the 81-cone triangulation and its 226 residue classes;

- exact equality of the conductor’s 812 required bases with the frozen list;
- sharp witness coverage of all 812 bases;
- explicit low-dimensional witnesses used by the conductor.

The core reproduction command is

```
pwsh -NoProfile -File research/fano-half-rank-profiles/verification/run_
all.ps1
```

It uses Python with SymPy and a C++17 compiler and writes binaries only below the operating system’s temporary directory. The source hashes are recorded in verification/SHA256SUMS.

The bounded check with $\max h \leq 6$ and the fixed-seed witness searches are regressions and construction tools. They are not substituted for the general nonrealizability arguments.

8 Relation to prior work and limits of the result

Pott, Schmidt, and Zhou [1] count pairs of alternating matrices according to the three ranks of $(A, B, A + B)$. This is the two-dimensional, three-point precursor of the present seven-point profile. Their result does not give the compatibility semigroup or minimum effective dimension for a three-dimensional parameter space.

Eisenbud and Koh [2] classify three-dimensional spaces of 4×4 and 5×5 alternating matrices over an algebraically closed field. Gow [3] studies rank-related dimension bounds for subspaces of bilinear forms, and Qiao [4] enumerates alternating-matrix spaces over finite fields. The present result does not refine those works as an orbit classification. Instead, it forgets the isomorphism type and retains the seven labelled ranks, allowing arbitrary ambient dimension over $\mathbb{F}_2$.

The following statements are outside the scope of the paper.

1. We do not classify alternating nets up to equivalence or count the nets with a given profile.
2. We do not treat arbitrary finite fields or odd characteristic.
3. We do not classify affine linear terms, Walsh signs, common zero-fibre histograms, or supports of three-dimensional subcodes of second-order Reed–Muller codes.
4. We do not infer formal peer-review status or global priority from the current literature search. The comparison above records the nearest public sources checked for this draft.

9 Conclusion

The seven half-ranks of a three-parameter alternating net over $\mathbb{F}_2$ are governed by a nearly saturated Fano triangle semigroup. Its failure of saturation has exactly two forms: primitive line jumps and singleton-face parity cosets. After realizability is settled, the minimum effective dimension is even and sharp except for a complete list of finite low-height orbits and one cut-generated infinite family.

The result illustrates a useful division of labor. Pfaffian and idempotent arguments explain the infinite obstructions, while finite exact polyhedral certificates close the additive sufficiency problem. Neither bounded enumeration nor unsuccessful search is used to assert a general theorem.

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